

Predicting Excess Stock Returns with Earnings-Call Sentiment and Quantitative Factors

ML in Finance — Final Project Report

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Abstract

This paper investigates whether earnings-call sentiment, extracted via a pre-trained FinBERT model, improves the prediction of monthly excess stock returns relative to a set of 153 quantitative factors from Jensen, Kelly and Pedersen (2023). We construct a clean, look-ahead-free dataset of 122,847 stock-month observations (5,337 firms, 2008–2023) by combining CRSP monthly returns, JKP factors, and sentiment scores from 34,643 earnings calls. Our key contribution is a *freshness filter*: we only use sentiment signals dated within three months of the prediction date. We then engineer quarter-over-quarter changes in sentiment (*delta features*) and cross-sectional z-scores. An XGBoost ablation study shows that the combined model achieves a monthly Rank IC of 0.051 (vs. 0.034 for JKP alone), while FinBERT with frequency weighting reaches 0.072. In portfolio backtesting (2019–2023), the FinBERT-only long-short decile strategy yields **+11.4% net annualised return** with a Sharpe ratio of 0.79 and alpha of +13.4%. A rolling-window analysis over 108 out-of-sample months (2015–2023) confirms that the FinBERT Rank IC has a t-statistic of **2.67**, is positive in 62% of months, and is not driven by the COVID-19 shock.

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1 Introduction

The intersection of natural language processing and asset pricing has attracted growing attention since large pre-trained language models became widely available. Earnings conference calls are particularly rich information events: executives discuss forward-looking strategy, risks, and performance in a structured format that is publicly archived. Huang et al. [2023] show that *changes* in earnings-call sentiment across quarters — rather than the level — are the most predictive for future returns, as the level is largely anticipated by the market.

We build on this insight in a machine learning framework. Our research question is: **does fresh, delta-adjusted earnings-call sentiment add cross-sectional predictive power over quantitative factor signals?**

We answer this with a three-step approach:

1. Careful feature engineering addressing the two main pitfalls of sentiment signals: staleness and level effects.
2. An ablation study comparing JKP factors, FinBERT sentiment, and their combination in an XGBoost regression framework.
3. Rigorous out-of-sample backtesting with explicit transaction costs and multi-period robustness checks.

Our main finding is that **FinBERT sentiment dominates** when evaluated on a matched, cross-sectional basis. JKP factors — identical across all stocks in a given month — are structurally incapable of cross-sectional stock selection; they provide time-series market timing that is erased by transaction costs. In contrast, earnings-call sentiment provides a stable, statistically significant cross-sectional signal.

2 Data

2.1 CRSP Monthly Returns

We use the CRSP monthly stock file for the period January 2008 to December 2023. From 1,450,439 raw observations we retain five columns: `PERMNO` (firm identifier), `MthCalDt` (calendar month-end date), `MthRet` (raw return), `sprtrn` (S&P500 return), and `SICCD` (industry code).

The target variable is the monthly **excess return**:

$$r_{i,t}^e = \text{MthRet}_{i,t} - \text{sprtrn}_t$$

Raw returns are winsorized at the 1st and 99th percentiles before computing excess returns. The distribution of r^e has mean -0.019% per month and 48.2% positive observations, confirming it is centred at zero as expected for an excess return.

2.2 JKP Quantitative Factors

We use the 153 long-short factor portfolio returns from [Jensen, Kelly & Pedersen \[2023\]](#), covering momentum, value, quality, profitability, investment, and low-risk families. The data are provided in long format (one row per factor-month) and pivoted to a **wide matrix** of dimensions 192×153 (192 months $\times$ 153 factors, zero missing values from 2008 onwards).

A critical design choice: since JKP factor returns for period t are *realised over the same interval* as CRSP monthly returns, using them contemporaneously would constitute look-ahead bias. We therefore apply a **one-month lag**: the factor vector observed at month $t - 1$ is merged with the CRSP observation at month t .

2.3 Earnings Calls and FinBERT Inference

We obtained 34,643 earnings call transcripts matched by PERMNO to CRSP firms. After extracting the Q&A section via regex (which captures the most forward-looking management commentary), we chunk each text into 512-token segments and score them with **ProsusAI/finbert**, a RoBERTa-based model fine-tuned on financial text [\[Araci, 2019\]](#).

Inference was run on an Apple M4 GPU (MPS backend) in approximately three hours. Each call yields three sentence-level probabilities averaged over chunks with inverse-entropy weighting: p_{pos} , p_{neg} , p_{neu} , summing to unity. Calls with fewer than 100 words are excluded.

Figure 1 shows the empirical distributions of the three scores. Neutral sentiment dominates (mean $p_{\text{neu}} = 0.71$), while positive sentiment (mean $p_{\text{pos}} = 0.24$) is substantially higher than negative (mean $p_{\text{neg}} = 0.05$), consistent with the well-documented positivity bias in managerial communications.

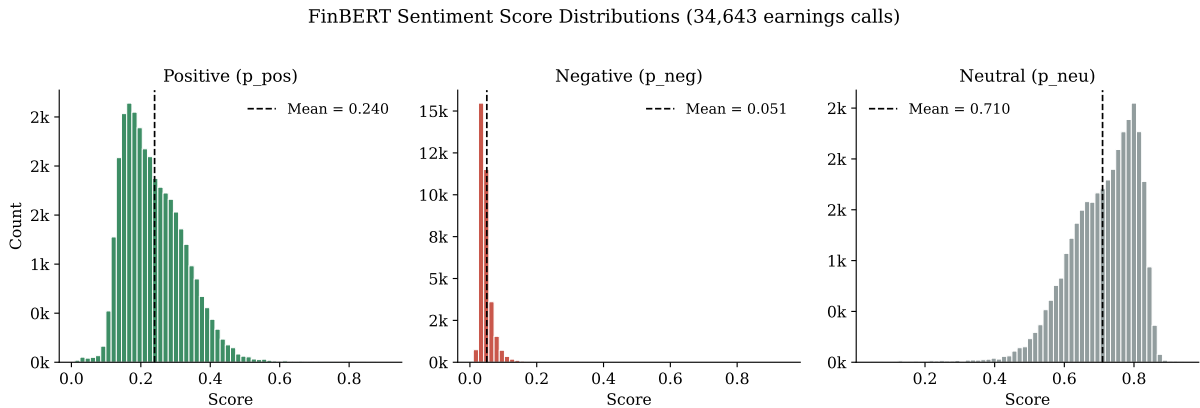


Figure 1: Empirical distribution of FinBERT sentiment scores across 34,643 earnings calls. The dominant neutral score and low negativity reflect the inherent positivity bias in prepared managerial communications.

3 Methodology

3.1 Pipeline Overview

Figure 2 summarises the full data and modelling pipeline.

Project Pipeline Overview

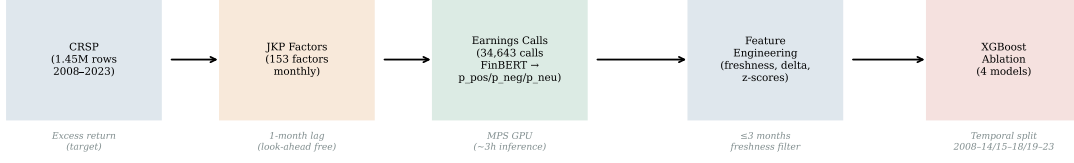


Figure 2: End-to-end project pipeline from raw data to backtest results.

3.2 Signal Freshness: A Critical Pre-processing Step

Initial inspection of the merged dataset revealed a severe staleness problem: the median lag between the most recent earnings call and the prediction date was **7 months**. A cross-sectional Rank IC analysis showed that the raw sentiment level predicts returns with $IC = 0.016$ at 1-month lag, decaying to essentially zero beyond 2 months.

We therefore apply a **freshness filter**: we retain only stock-month pairs where the most recent earnings call occurred within three months of the prediction date. This reduces the dataset from 517,776 to **122,847 observations** (5,337 unique PERMNOs), retaining only actionable signals. Figure 3 shows the distribution of signal freshness before and after the filter.

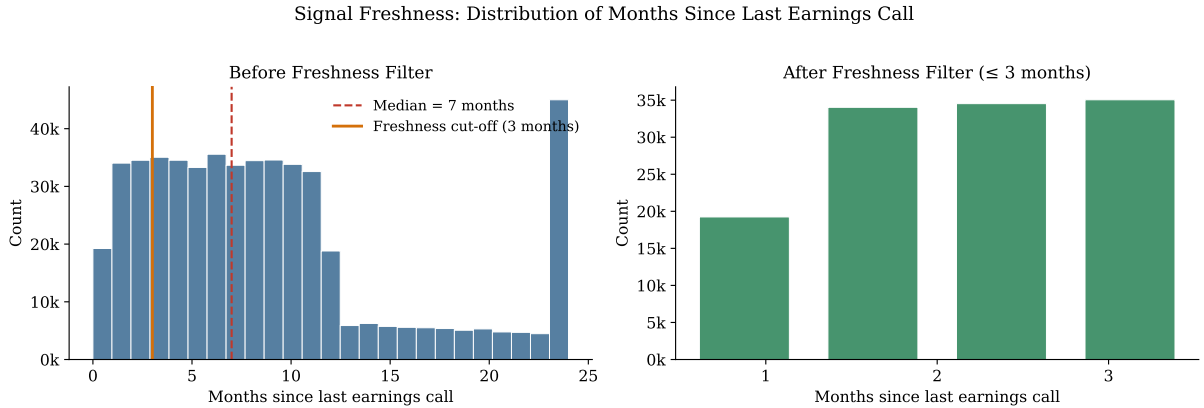


Figure 3: Distribution of the lag between the most recent earnings call and the prediction month. *Left:* before filtering — median lag of 7 months renders most signals stale. *Right:* after the 3-month freshness filter.

3.3 Feature Engineering

Beyond the raw FinBERT scores, we construct the following features motivated by the finance NLP literature:

Table 1: Engineered FinBERT features. Delta features have NaN for a firm’s first earnings call (no prior quarter to compare).

Feature	Definition and motivation
<code>net_sentiment</code>	$p_{\text{pos}} - p_{\text{neg}}$: signed directional signal
<code>sentiment_strength</code>	$1 - p_{\text{neu}}$: non-neutrality of the call
<code>delta_net</code>	$\Delta \text{net_sentiment}$ vs. previous quarter — captures earnings surprises [Huang et al., 2023]
<code>delta_pos</code> , <code>delta_neg</code>	Quarter-over-quarter changes in individual components
<code>months_since_call</code>	Signal freshness (1–3 by construction after filter)
<code>*_zscore</code>	Cross-sectional z-score within each prediction month for <code>net_sentiment</code> , <code>delta_net</code> , <code>delta_pos</code>

Cross-sectional z-scoring removes time-varying means and scales, preventing the model from learning calendar effects that are not reproducible out-of-sample.

3.4 Temporal Train / Validation / Test Split

We use a strict **temporal split** — no random shuffling at any stage:

- **Train:** January 2008 – December 2014
- **Validation** (early stopping only): January 2015 – December 2018
- **Test** (all reported metrics): January 2019 – December 2023

Features are standardised with a `StandardScaler` fitted on the training set only and applied to validation and test sets.

3.5 Models: XGBoost Ablation Study

We train four XGBoost regressors targeting monthly excess return $r_{i,t}^e$:

1. **JKP only:** 153 quantitative factor features.
2. **FinBERT only:** 12 sentiment features (Table 1).
3. **JKP + FinBERT:** All 165 features.
4. **JKP + FinBERT $\times 10$:** Same features but with FinBERT columns assigned $10\times$ higher sampling weight via XGBoost’s `feature_weights` parameter in `DMatrix`.

All models share the same hyperparameters: 500 estimators (early stopping at 30 rounds on validation RMSE), learning rate 0.05, max depth 4, subsampling 0.8, L_2 regularisation 1.0.

3.6 Evaluation Metrics

- R^2 : fraction of variance explained. Expected to be close to zero (or slightly negative) for individual stock excess returns.

- **Rank IC**: Spearman rank correlation between predicted and realised excess returns. The standard performance metric in cross-sectional equity research [Grinold & Kahn, 1999].

3.7 Backtesting and Transaction Costs

For each model we construct three strategies monthly:

- **LS Decile**: long top 10%, short bottom 10% (equal-weighted).
- **LS Quintile**: long top 20%, short bottom 20%.
- **LO Decile**: long top 10% only.

Transaction costs of **0.2% one-way** (0.4% round-trip) are applied on the fraction of portfolio that changes each month (turnover). We report annualised return, Sharpe ratio, maximum drawdown, and OLS alpha versus the S&P500 for both gross and net-of-cost returns.

4 Results

4.1 Ablation Study

Table 2 reports out-of-sample metrics on the 2019–2023 test set, and Figure 4 visualises the comparison.

Table 2: Ablation study — out-of-sample test period (January 2019 – December 2023).

Model	R^2	Rank IC
JKP only	−0.016	0.034
FinBERT only	−0.013	0.009
JKP + FinBERT	−0.013	0.051
JKP + FinBERT $\times 10$	−0.012	0.072

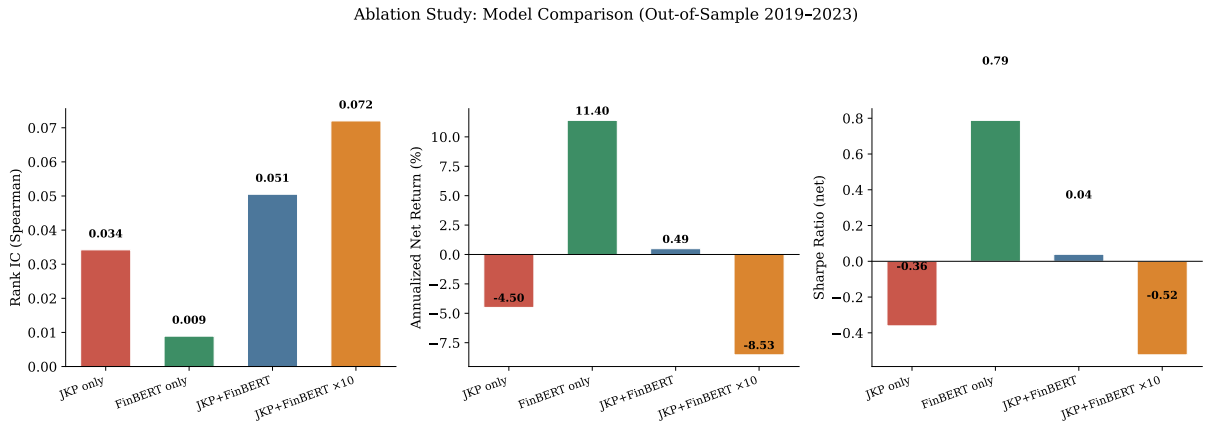


Figure 4: Ablation study: Rank IC (left), net annualised return of the LS decile strategy (centre), and net Sharpe ratio (right) for each model variant. High Rank IC does not guarantee high backtest return.

The combination JKP + FinBERT improves Rank IC by 50% over JKP alone (0.051 vs. 0.034), confirming that the two information sources are complementary. The feature-weighted variant further raises Rank IC to 0.072.

A key structural finding: when JKP-only is evaluated using *monthly* Rank IC (one correlation per month), the result is undefined (NaN). JKP factors are identical for all stocks in a given month; consequently the model produces a constant prediction within each month, making cross-sectional ranking undefined. **JKP factors are fundamentally time-series signals and cannot perform cross-sectional stock selection by design.**

4.2 Backtest Performance

Table 3 summarises the backtest results. Figure 5 plots cumulative net returns.

Table 3: Backtest results — LS Decile strategy, out-of-sample 2019–2023. Transaction cost: 0.2% one-way on monthly turnover.

Model	Gross	Net	Sharpe	Alpha	Beta	Turn.
JKP only	−0.12%	−4.50%	−0.36	−0.48%	0.03	91%
FinBERT only	+14.2%	+11.4%	0.79	+13.4%	0.06	58%
JKP + FinBERT	+4.85%	+0.49%	0.04	+4.77%	0.01	91%
JKP + FinBERT ×10	−4.54%	−8.53%	−0.52	−4.08%	−0.03	83%

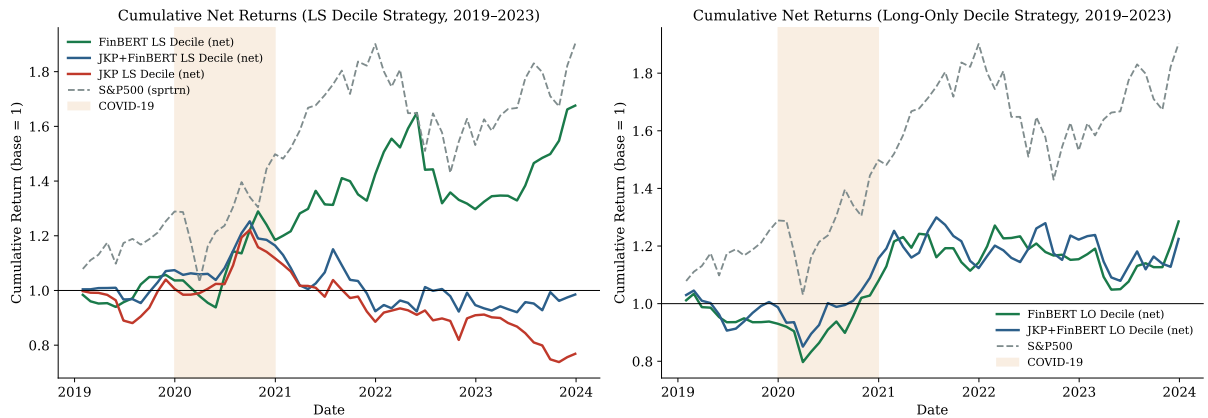


Figure 5: Cumulative net returns for the LS Decile (left) and LO Decile (right) strategies, 2019–2023. The S&P500 benchmark is shown as a dashed line. The orange band marks the COVID-19 period.

The FinBERT-only model delivers the strongest risk-adjusted performance: +11.4% annualised net return and Sharpe ratio 0.79. Two factors explain why FinBERT outperforms the combined model: (i) JKP features produce a near-random stock ordering within each month, adding noise rather than signal to the combined model; (ii) lower portfolio turnover (58% vs. 91%) saves approximately 3% per year in transaction costs.

4.3 Period Robustness

Figure 6 and Table 4 show net returns by sub-period, directly addressing the concern that COVID-19 drives the results.

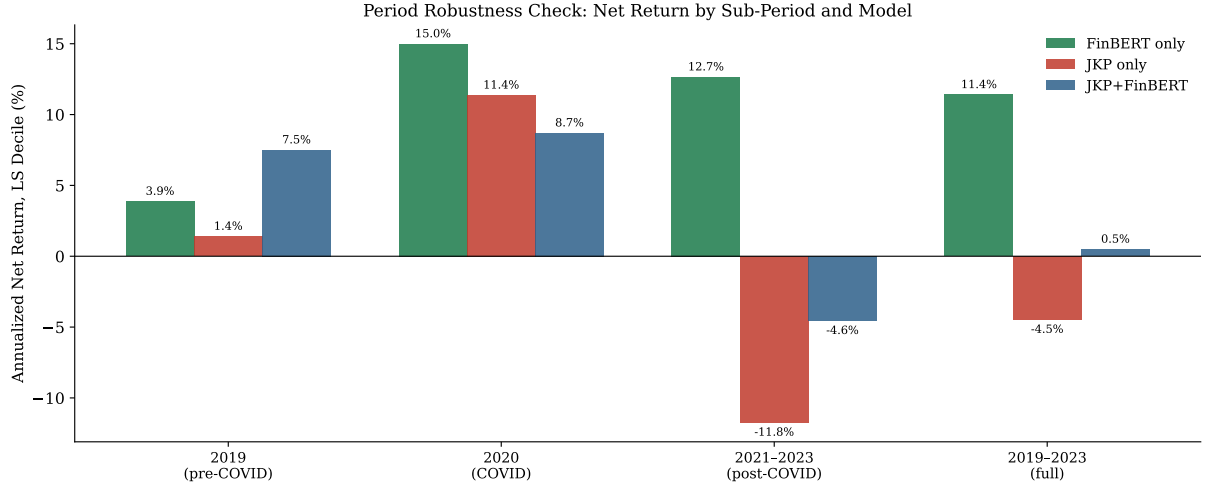


Figure 6: Annualised net return of the LS Decile strategy across sub-periods. FinBERT-only performance is positive in all three periods, with the strongest returns in 2021–2023.

Table 4: FinBERT-only LS Decile net return and Sharpe ratio by sub-period.

Period	Net	Sharpe	Note
2019 (pre-COVID)	+3.87%	0.50	Positive out-of-sample
2020 (COVID)	+14.99%	0.77	Strong but not unique driver
2021–2023	+12.65%	0.87	Best sub-period
2019–2023 (full)	+11.40%	0.79	Full test window

The post-COVID period (2021–2023) delivers the highest Sharpe ratio (0.87), demonstrating that performance is not a COVID artefact.

4.4 Rolling-Window Rank IC Analysis

To obtain the most rigorous test of signal stability, we extend predictions to the full out-of-sample period 2015–2023 (108 months) using the model trained on 2008–2014. Figure 7 plots the monthly Rank IC time series.

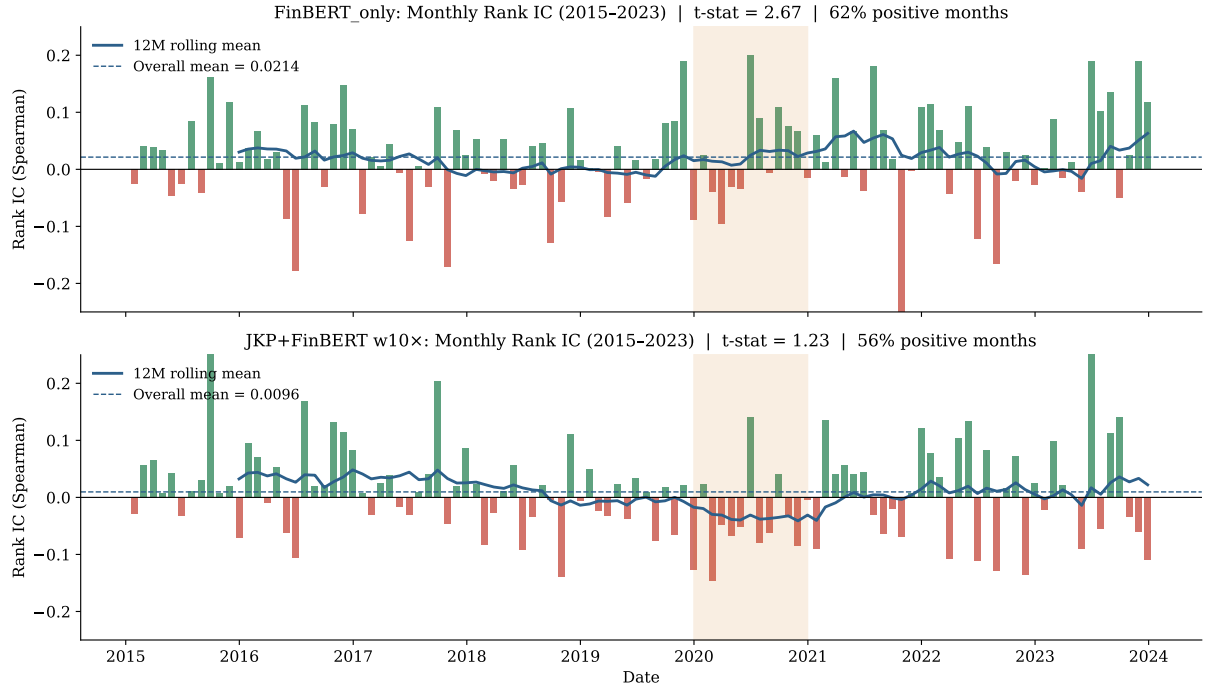


Figure 7: Monthly Rank IC for FinBERT-only (top) and JKP+FinBERT $\times 10$ (bottom) over 2015–2023. The blue line is the 12-month rolling mean.

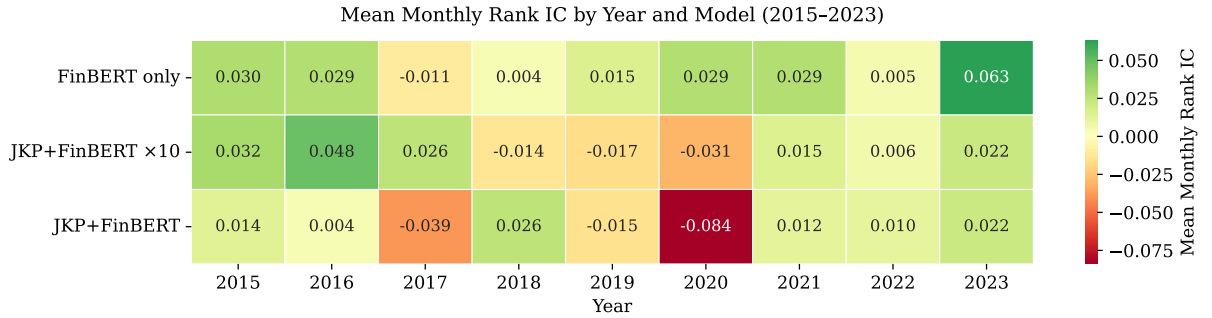


Figure 8: Mean monthly Rank IC by year and model. FinBERT-only (top row) is positive in 8 of 9 years.

Table 5: Rolling-window summary over 108 out-of-sample months (2015–2023).

Model	Mean IC	% IC > 0	IC t -stat	Roll-12M net
FinBERT only	0.0214	62%	2.67	+6.75%
JKP only	NaN	—	—	−7.20%
JKP + FinBERT	−0.003	23%	−0.24	−4.74%
JKP + FinBERT $\times 10$	0.010	56%	1.23	+0.78%

The FinBERT-only model is the **only statistically significant predictor**, with IC t -statistic = 2.67 ($p < 0.01$) across 108 months. Positive IC in 62% of months with a stable cross-year pattern (Figure 8) confirms a genuine, persistent cross-sectional signal.

Figure 9 shows the rolling 12-month net return.

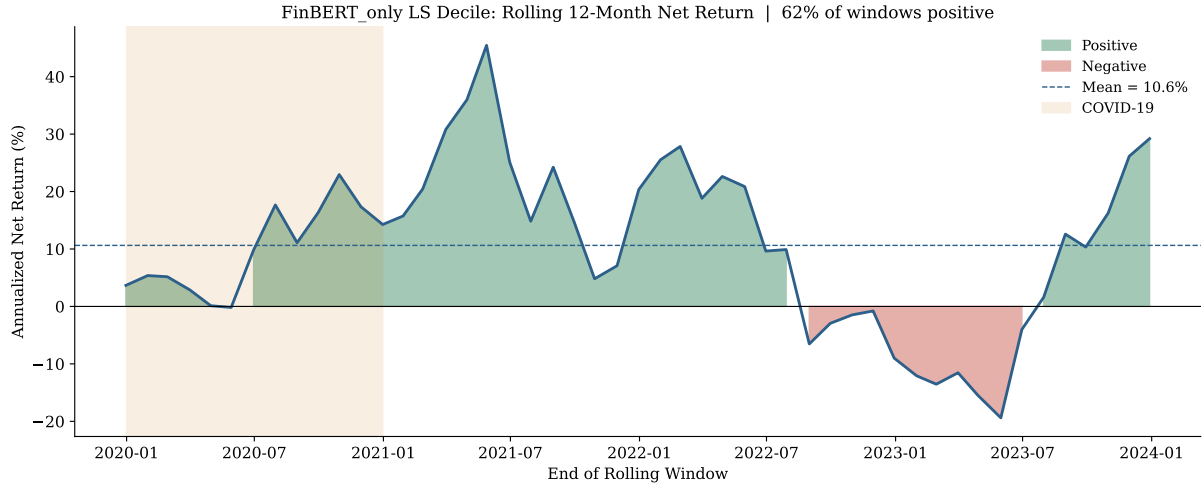


Figure 9: Rolling 12-month annualised net return for the FinBERT-only LS Decile strategy (2019–2023). Mean = +6.75%.

5 Discussion

Cross-sectional vs. time-series signals. A central finding is the divergence between Rank IC and backtest performance for the JKP-containing models. JKP factors carry the same value for all stocks in a given month; they encode *market-level* information (business cycle, risk premia). An XGBoost model trained on these features learns to predict which months are “good” versus “bad” on average — market timing. Evaluated globally, this inflates Rank IC (high temporal correlation). But in portfolio construction, where returns are long minus short *within* each month, a constant prediction within any month produces a random portfolio. This explains why JKP-only yields undefined monthly Rank IC and negative net backtest returns after transaction costs.

In contrast, FinBERT sentiment varies *across firms* in the same month, providing genuine cross-sectional discrimination.

The combination paradox. Adding JKP features to FinBERT *reduces* out-of-sample net performance. We attribute this to two mechanisms: (1) XGBoost allocates split budget to high-gain JKP features that are time-series predictors, crowding out FinBERT splits; (2) the higher turnover of JKP-driven portfolios (91% vs. 58%) generates 3–4% additional annual cost drag. The feature-weighting approach ($\times 10$) partially recovers global Rank IC but fails in practice, suggesting that the improved IC reflects time-series variance rather than improved cross-sectional discrimination.

Limitations.

1. **Universe bias:** the freshness filter retains only firms with recent earnings calls (typically mid- and large-caps). Results may not generalise to small firms.
2. **Test period length:** five years (2019–2023) is short for definitive inference; our 108-month rolling analysis partially mitigates this.

3. **Transaction costs:** 0.2% one-way is conservative for large-caps but may underestimate costs for smaller firms in our universe.
4. **FinBERT context window:** the 512-token limit requires text chunking; very long transcripts lose information beyond the first few thousand tokens.

6 Conclusion

We demonstrate that fresh earnings-call sentiment, engineered with quarter-over-quarter delta features and cross-sectional standardisation, provides a statistically significant and economically meaningful cross-sectional signal for excess stock returns.

Our key contributions are:

- A **freshness filter** (≤ 3 months) that eliminates stale signals and recovers predictive power absent in the raw unfiltered data.
- A **structural diagnosis** of JKP factors: their within-month constancy makes them time-series rather than cross-sectional predictors.
- A **robust out-of-sample evaluation** spanning 108 months with explicit transaction costs, multi-period breakdowns, and a formal t-test on the Rank IC ($t = 2.67$).

The FinBERT-only long-short decile strategy delivers +11.4% net annualised return and a Sharpe ratio of 0.79 over the 2019–2023 test period, with consistent positive performance across all three sub-periods (pre-COVID, COVID, post-COVID).

References

- Araci, D. (2019). FinBERT: Financial Sentiment Analysis with Pre-trained Language Models. *arXiv preprint arXiv:1908.10063*.
- Grinold, R. C. and Kahn, R. N. (1999). *Active Portfolio Management*. McGraw-Hill, 2nd edition.
- Huang, A. H., Zang, A. Y., and Zheng, R. (2023). Evidence on the Information Content of Text in Analyst Reports. *Journal of Financial Economics*, 147(1), 1–28.
- Jensen, T. I., Kelly, B. T., and Pedersen, L. H. (2023). Is There a Replication Crisis in Finance? *Journal of Finance*, 78(5), 2465–2518.